Toric varieties

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Lecture 1

In this course, we do everything over $\mathbb{C}$.

1 What is a toric variety?

Definition 1.1 (affine variety)
An *affine variety* is $V = \mathrm{Spec}(R)$, where R is a finitely generated integral[a] $\mathbb{C}$-algebra.

[a] R is an integral domain.

Fact: Given such an R, we can choose generators and relations, and write

$$R = \frac{\mathbb{C}[x_1, \ldots, x_n]}{I}$$

where I is a prime ideal. This gives an affine embedding. i.e. a closed embedding

$$V = \mathbb{V}(I) = \mathbb{A}^n$$

Note that the affine embedding is not unique.

Definition 1.2 ((complex) variety)
A *(complex) variety* is a scheme X which is integral (reduced and irreducible), separated, and of finite type.

We can avoid schemes, and glue varieties out of affine varieties. We need

1. a finite collection $\{V_\alpha\}_{\alpha \in A}$ of affine varieties, say $V_\alpha = \mathrm{Spec}(R_\alpha)$,
2. for $\alpha, \beta \in A$, we need Zariski opens $V_{\alpha\beta} \subseteq V_\alpha$,
3. algebraic transition maps $g_{\alpha\beta} : V_{\alpha\beta} \to V_{\beta\alpha}$, satisfying the usual cocycle conditions,
4. check that the resulting
$$X = \frac{\bigsqcup_\alpha V_\alpha}{\sim}$$
is irreducible and separated (Hausdorff).

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Definition 1.3 (torus)
An *(algebraic complex) torus of dimension n* is

$$\mathrm{T}^n = (\mathbb{C}^*)^n = \mathrm{Spec}(\mathbb{C}[x_1^{\pm 1}, \ldots, x_n^{\pm 1}]) = (\mathbb{A}^1 \setminus \{0\})^n$$

Note that this is not the same as the usual torus $(S^1)^n$. In particular, we have an affine embedding

$$\mathrm{T}^n = \mathrm{Spec}\left(\frac{\mathbb{C}[z_0, \ldots, z_n]}{\langle z_0 \cdots z_n - 1\rangle}\right) = \mathbb{V}(z_0 \cdots z_n - 1) \subseteq \mathbb{A}^{n+1}$$

Exercise: Check that we have an isomorphism of rings

$$\mathbb{C}[z_1^{\pm 1}, \ldots, z_n^{\pm 1}] \cong \frac{\mathbb{C}[z_0, \ldots, z_n]}{\langle z_0 \cdots z_n - 1\rangle}$$

The torus is also a *commutative group scheme*, that is, we have algebraic morphisms

$$\begin{aligned} e &: \bullet \to \mathrm{T}^n \\ m &: \mathrm{T}^n \times \mathrm{T}^n \to \mathrm{T}^n \\ i &: \mathrm{T}^n \to \mathrm{T}^n \end{aligned}$$

making T^n into an abelian group with identity e, multiplication m and inverse i. On the level of coordinate rings, we have the *identity section*,

$$\begin{aligned} e^\sharp : \mathbb{C}[t_1^{\pm 1}, \ldots, t_n^{\pm 1}] &\to \mathbb{C} \\ t_i^{\pm 1} &\mapsto 1 \end{aligned}$$

The multiplication map is given by pointwise operations, i.e.

$$m((x_1, \ldots, x_n), (y_1, \ldots, y_n)) = (x_1 y_1, \ldots, x_n y_n)$$

With this, we have *comultiplication*

$$\begin{aligned} m^\sharp : \mathbb{C}[t_1^{\pm 1}, \ldots, t_n^{\pm 1}] &\to \mathbb{C}[x_1^{\pm 1}, \ldots, x_n^{\pm 1}] \otimes \mathbb{C}[y_1^{\pm 1}, \ldots, y_n^{\pm 1}] \\ t_i &\mapsto x_i \otimes y_i \end{aligned}$$

and the inverse map is given by

$$\begin{aligned} i^\sharp : \mathbb{C}[t_1^{\pm 1}, \ldots, t_n^{\pm 1}] &\to \mathbb{C}[t_1^{\pm 1}, \ldots, t_n^{\pm 1}] \\ t_i &\mapsto t_i^{-1} \end{aligned}$$

Definition 1.4 (toric variety)
A *toric variety* is a variety X with a dense open $\mathrm{T}^n \subseteq X$, and an action (i.e. algebraic map)

$$a : \mathrm{T}^n \times X \to X$$

which extends the multiplication on X.

Example 1.5
T^n is itself a toric variety.

Cohomology

Example 1.6

$\mathbb{A}^n = \mathrm{Spec}(\mathbb{C}[x_1, \ldots, x_n])$, with the torus

$$\mathrm{Spec}(\mathbb{C}[x_1^{\pm 1}, \ldots, x_n^{\pm 1}])$$

with

$$a((t_1, \ldots, t_n), (x_1, \ldots, x_n)) = (t_1 x_1, \ldots, t_n x_n)$$

Example 1.7

Projective space

$$\mathbb{P}^n = \mathrm{Proj}(\mathbb{C}[z_0, \ldots, z_n]) = \{(z_0 : \cdots : z_n) \mid (z_0, \ldots, z_n) \neq (0, \ldots, 0)\}$$

is also a toric variety, with torus

$$\mathrm{T}^n = \{(z_0 : \cdots : z_n) \mid z_0 \cdots z_n = 1\}$$

with action given coordinate-wise again, i.e.

$$a((t_0 : \cdots : t_n), (z_0 : \cdots : z_n)) = (t_0 z_0 : \cdots : t_n z_n)$$

in non-homogeneous coordinates (i.e. in the affine $z_0 \neq 0$) of the torus, we can define the action by

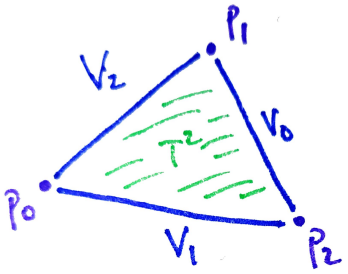
$$(t_1, \ldots, t_n) \cdot (z_0 : \cdots : z_n) = (z_0 : t_1 z_1 : \cdots : t_n z_n)$$

We can check that T^n is contained in all of the standard opens $U_i = D(z_i)$, and if we choose the usual affine coordinates, then T^n embeds into affine space as above.

Toric varieties are determined by the dimension n of the torus, and the *toric boundary* $X \setminus \mathrm{T}^n$. For example, $\mathbb{P}^2 \setminus \mathrm{T}^2$ has three components,

1. $V_0 = \mathbb{V}(z_0)$,
2. $V_1 = \mathbb{V}(z_1)$,
3. $V_2 = \mathbb{V}(z_2)$

The V_i intersect pairwise, but not all three at the same time. Each V_i is invariant under the action of the torus, so $\mathbb{P}^2$ can be represented as



In particular, this gives us a stratification of $\mathbb{P}^2$, with strata T^2, $\mathbb{C}^* = T^1$ corresponding to V_i with the vertices removed, and the three vertices.

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